Throughput QOS

2026-02-23-12-22-54

Objective

The objective of the QoS (Quality of Service) traffic throughput test is to measure the maximum achievable throughput of a network under specific QoS settings and conditions.By conducting this test, we aim to assess the capacity of network to handle high volumes of traffic while maintaining acceptable performance levels,ensuring that the network meets the required QoS standards and can adequately support the expected user demands.

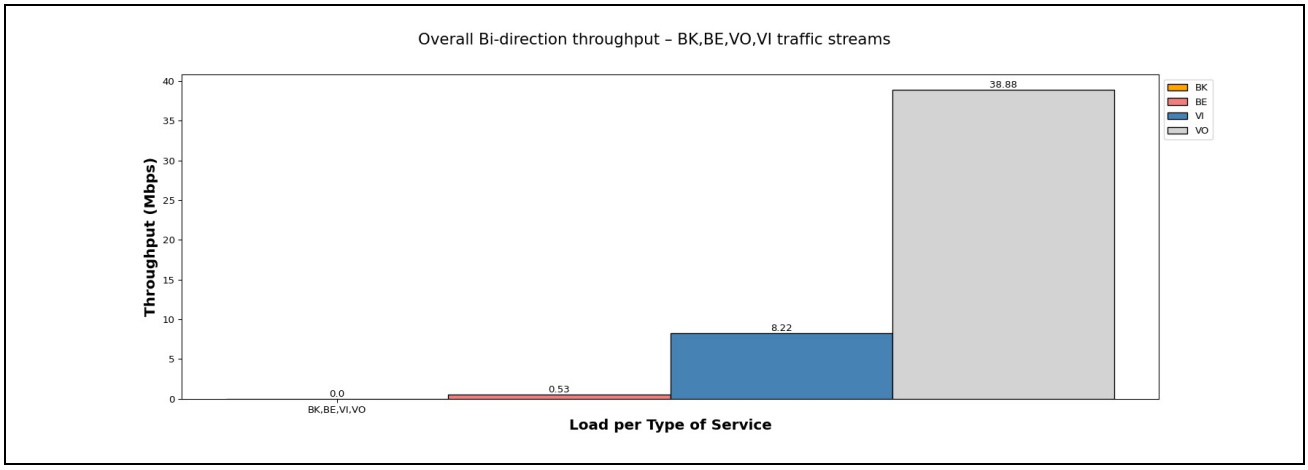
Test Configuration	Number of Stations	4
	AP Model	Test_DUT
	SSID_2.4GHz	NETGEAR_2G_wpa2
	SSID_5GHz	
	SSID_6GHz	
	Traffic Duration in hours	0.02
	Security_2.4GHz	wpa2
	Security_5GHz	
	Security_6GHz	
	Protocol	TCP
	Traffic Direction	Bi-direction
	TOS	['VI', 'VO', 'BK', 'BE']
	Per TOS Load in Mbps	Upload:100.0 Mbps,Download:100.0 Mbps

Overall Bi-direction throughput for all TOS i.e BK | BE | Video (VI) | Voice (VO)

No of Stations	Throughput for Load 100.0 Mbps-upload	Throughput for Load 100.0 Mbps-download
4	BK : 0.0, BE : 0.26, VI: 2.32, VO: 1.21	BK : 0.0, BE : 0.27, VI: 5.9, VO: 37.67

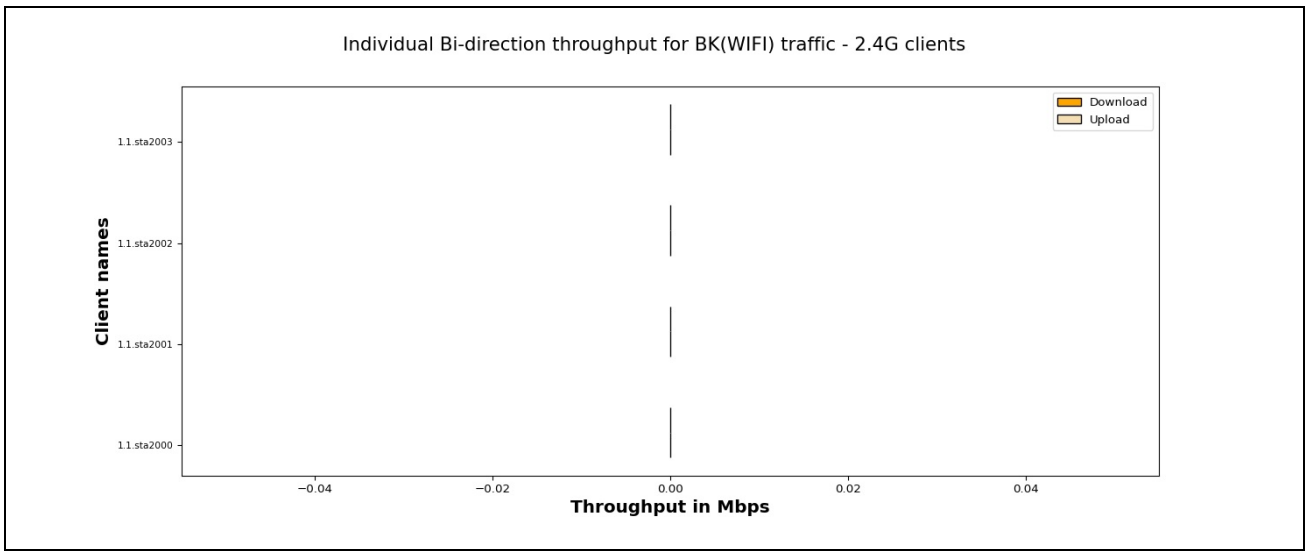
Overall Bi-direction throughput for 4 clients for 100.0 Mbps-download band with different TOS.

The below graph represents overall Bi-direction throughput for all connected stations running BK, BE, VO, VI traffic with different intended loads per station – 100.0 Mbps-download



Individual Bi-direction throughput with intended load Upload:100.0 Mbps,Download:100.0 Mbps/station for traffic BK(WiFi).

The below graph represents individual throughput for 4 clients running BK (WiFi) traffic. Y- axis shows “Client names“ and X-axis shows “Throughput in Mbps”.

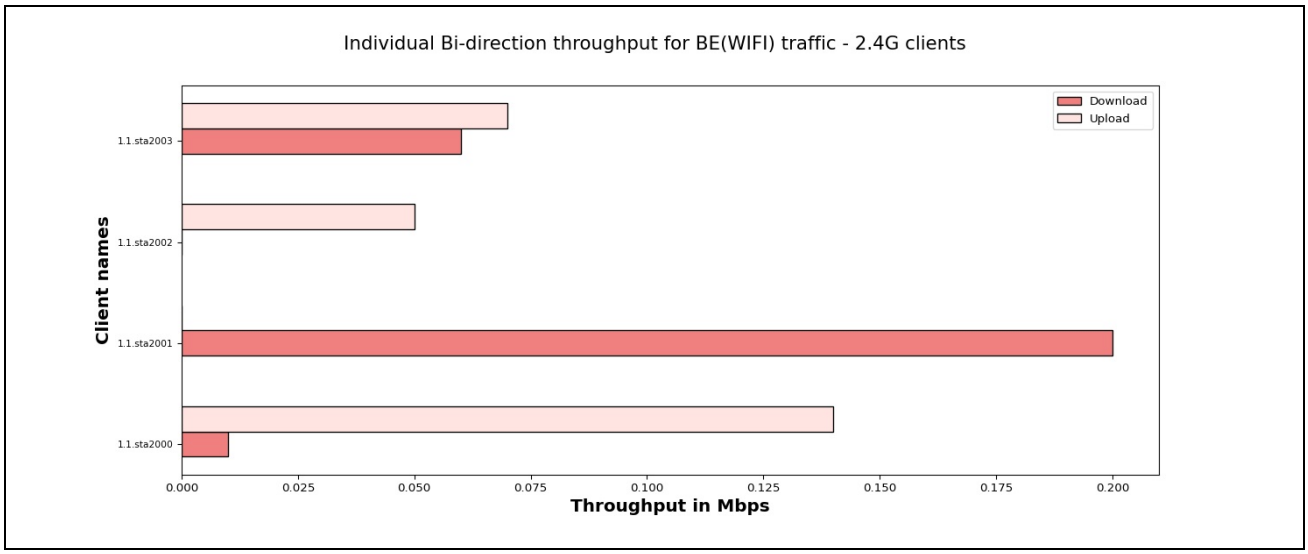


TOS : Background

Client Name	Mac	Channel	SSID	Type of traffic	Traffic Direction	Traffic Protocol	Offered upload rate(Mbps)	Offered download rate(Mbps)	Observed upload rate(Mbps)	Observed download rate(Mbps)	Observed Upload Drop (%)	Observed Download Drop (%)
1.1.sta2000	04:f0:21:89:e7:f4	11	NETGEAR_2G_wpa2	Background	Bi-direction	TCP	100.0 Mbps	100.0 Mbps	0.0	0.0	100.0	100.0
1.1.sta2001	04:f0:21:89:1a:f4	11	NETGEAR_2G_wpa2	Background	Bi-direction	TCP	100.0 Mbps	100.0 Mbps	0.0	0.0	100.0	100.0
1.1.sta2002	04:f0:21:89:20:f4	11	NETGEAR_2G_wpa2	Background	Bi-direction	TCP	100.0 Mbps	100.0 Mbps	0.0	0.0	100.0	100.0
1.1.sta2003	04:f0:21:89:7e:f4	11	NETGEAR_2G_wpa2	Background	Bi-direction	TCP	100.0 Mbps	100.0 Mbps	0.0	0.0	75.0	100.0

Individual Bi-direction throughput with intended load Upload:100.0 Mbps,Download:100.0 Mbps/station for traffic BE(WiFi).

The below graph represents individual throughput for 4 clients running BE (WiFi) traffic. Y- axis shows "Client names" and X-axis shows "Throughput in Mbps".

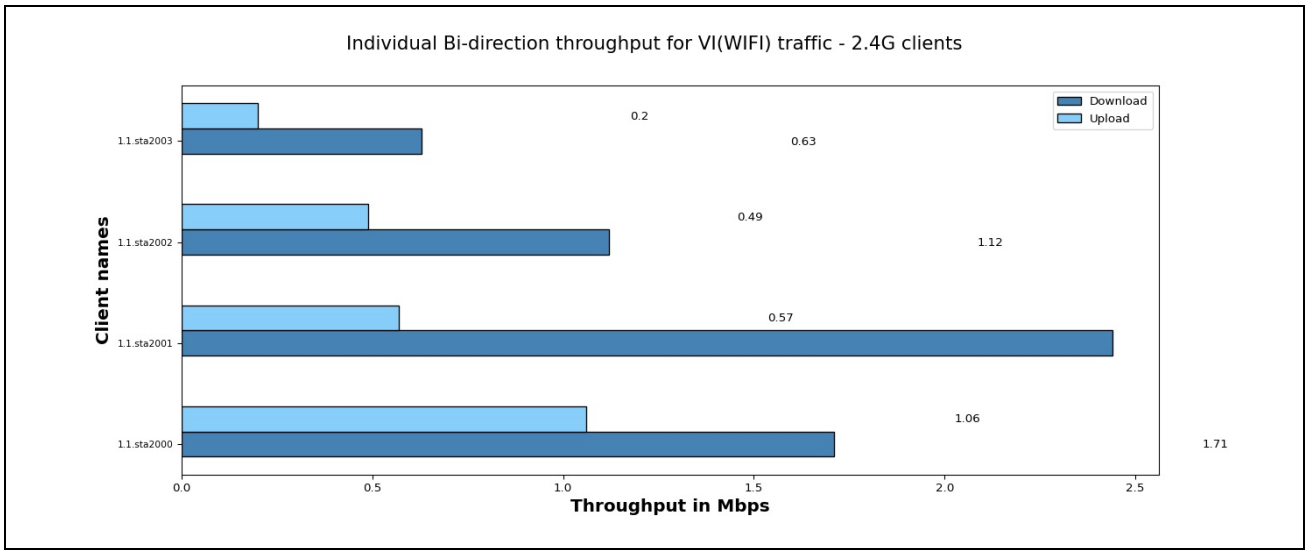


TOS : Besteffort

Client Name	Mac	Channel	SSID	Type of traffic	Traffic Direction	Traffic Protocol	Offered upload rate(Mbps)	Offered download rate(Mbps)	Observed upload rate(Mbps)	Observed download rate(Mbps)	Observed Upload Drop (%)	Observed Download Drop (%)
1.1.sta2000	04:f0:21:89:e7:f4	11	NETGEAR_2G_wpa2	Besteffort	Bi-direction	TCP	100.0 Mbps	100.0 Mbps	0.14	0.01	45.454545	33.333333
1.1.sta2001	04:f0:21:89:1a:f4	11	NETGEAR_2G_wpa2	Besteffort	Bi-direction	TCP	100.0 Mbps	100.0 Mbps	0.00	0.20	27.272727	73.333333
1.1.sta2002	04:f0:21:89:20:f4	11	NETGEAR_2G_wpa2	Besteffort	Bi-direction	TCP	100.0 Mbps	100.0 Mbps	0.05	0.00	41.666667	23.529412
1.1.sta2003	04:f0:21:89:7e:f4	11	NETGEAR_2G_wpa2	Besteffort	Bi-direction	TCP	100.0 Mbps	100.0 Mbps	0.07	0.06	57.142857	50.000000

Individual Bi-direction throughput with intended load Upload:100.0 Mbps,Download:100.0 Mbps/station for traffic VI(WiFi).

The below graph represents individual throughput for 4 clients running VI (WiFi) traffic. Y- axis shows "Client names" and X-axis shows "Throughput in Mbps".

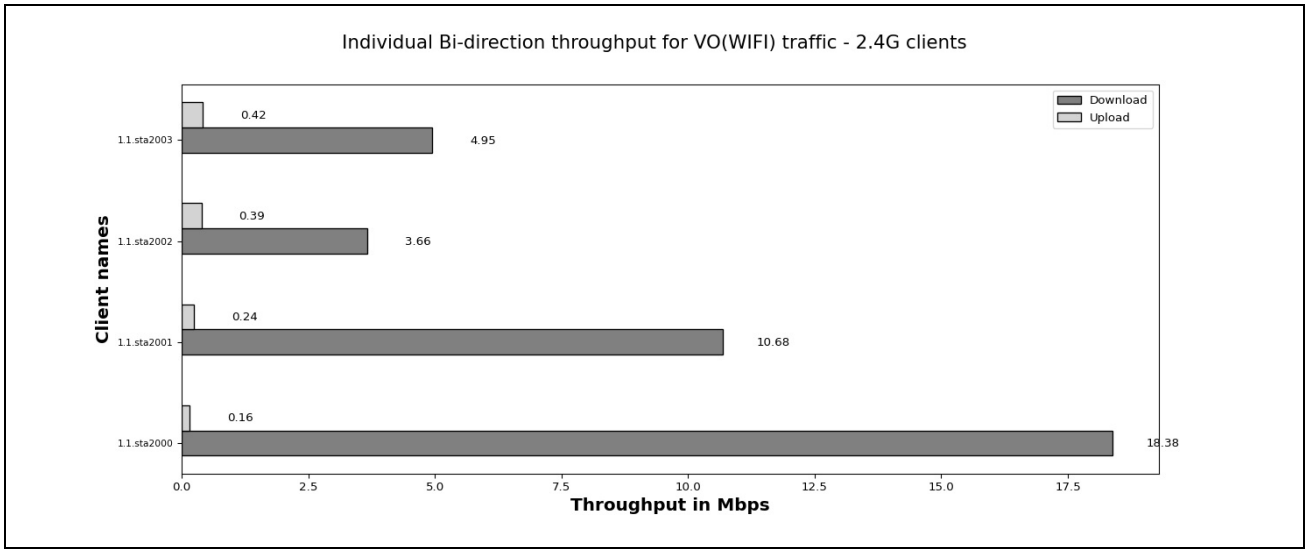


TOS : Video

Client Name	Mac	Channel	SSID	Type of traffic	Traffic Direction	Traffic Protocol	Offered upload rate(Mbps)	Offered download rate(Mbps)	Observed upload rate(Mbps)	Observed download rate(Mbps)	Observed Upload Drop (%)	Observed Download Drop (%)
1.1.sta2000	04:f0:21:89:e7:f4	11	NETGEAR_2G_wpa2	Video	Bi-direction	TCP	100.0 Mbps	100.0 Mbps	1.06	1.71	29.319372	25.423729
1.1.sta2001	04:f0:21:89:1a:f4	11	NETGEAR_2G_wpa2	Video	Bi-direction	TCP	100.0 Mbps	100.0 Mbps	0.57	2.44	14.563107	11.872146
1.1.sta2002	04:f0:21:89:20:f4	11	NETGEAR_2G_wpa2	Video	Bi-direction	TCP	100.0 Mbps	100.0 Mbps	0.49	1.12	25.252525	21.965318
1.1.sta2003	04:f0:21:89:7e:f4	11	NETGEAR_2G_wpa2	Video	Bi-direction	TCP	100.0 Mbps	100.0 Mbps	0.20	0.63	15.789474	24.050633

Individual Bi-direction throughput with intended load Upload:100.0 Mbps,Download:100.0 Mbps/station for traffic VO(WiFi).

The below graph represents individual throughput for 4 clients running VO (WiFi) traffic. Y- axis shows "Client names" and X-axis shows "Throughput in Mbps".



TOS : Voice

Client Name	Mac	Channel	SSID	Type of traffic	Traffic Direction	Traffic Protocol	Offered upload rate(Mbps)	Offered download rate(Mbps)	Observed upload rate(Mbps)	Observed download rate(Mbps)	Observed Upload Drop (%)	Observed Download Drop (%)
1.1.sta2000	04:f0:21:89:e7:f4	11	NETGEAR_2G_wpa2	Voice	Bi-direction	TCP	100.0 Mbps	100.0 Mbps	0.16	18.38	26.470588	13.766389
1.1.sta2001	04:f0:21:89:1a:f4	11	NETGEAR_2G_wpa2	Voice	Bi-direction	TCP	100.0 Mbps	100.0 Mbps	0.24	10.68	21.153846	1.702128
1.1.sta2002	04:f0:21:89:20:f4	11	NETGEAR_2G_wpa2	Voice	Bi-direction	TCP	100.0 Mbps	100.0 Mbps	0.39	3.66	15.151515	30.793651

1.1.sta2003	04:f0:21:89:7e:f4	11	NETGEAR_2G_wpa2	Voice	Bi-direction	TCP	100.0 Mbps	100.0 Mbps	0.42	4.95	9.523810	10.110803

Information	contact	support@candelatech.com